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# The genome of *Symbiodiniaceae*-associated *Stutzerimonas frequens* CAM01 reveals a broad spectrum of antibiotic resistance genes indicating anthropogenic drift in the Palk Bay coral reef of south-eastern India

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## Abstract

An increase in antibiotic pollution in reef areas will lead to the emergence of antibiotic-resistant bacteria, leading to ecological disturbances in the sensitive coral holobiont. This study provides insights into the genome of antibiotics-resistant *Stutzerimonas frequens* CAM01, isolated from *Favites*-associated *Symbiodiniaceae* of a near-shore polluted reef of Palk Bay, India. The draft genome contains 4.67 Mbp in size with 52 contigs. Further genome analysis revealed the presence of four antibiotic-resistant genes, namely, *adeF*, *rsmA*, APH (3")-Ib, and APH (6)-Id that provide resistance by encoding resistance–modulation–cell division (RND) antibiotic efflux pump and aminoglycoside phosphotransferase. The isolate showed resistance against 73% of the antibiotics tested, concurrent with the predicted AMR genes. Four secondary metabolites, namely Aryl polyene, NRPS-independent-siderophore, terpenes, and ectoine were detected in the isolate, which may play a role in virulence and pathogenicity adaptation in microbes. This study provides key insights into the genome of *Stutzerimonas frequens* CAM01 and highlights the emergence of antibiotic-resistant bacteria in coral reef ecosystems.

**Keywords** Antibiotic resistance · Coral reefs · Palk Bay · *Stutzerimonas frequens* · Antibiotic-resistant genes

## Introduction

Antibiotic resistance (AR) has emerged as a major global public health crisis of the twenty-first century (WHO 2014; Pruden et al. 2006; Seethalakshmi et al. 2021). Over the decades, excessive consumption and improper disposal of antibiotics have posed several threats to the ecosystem and human health (Yang et al. 2018). It is estimated that in the absence of a stringent policy, global antibiotic consumption would be 200% higher by the year 2030, as compared to 2015 (Klein et al. 2018). The unprecedented use of

antibiotics puts the microorganisms under selective pressure, leading to the enrichment of antibiotic-resistant genes, thus driving evolution of antibiotic resistance (Wang et al. 2019a, b). Antibiotics and their metabolites have spread across freshwater and marine ecosystems as noted by their detection in groundwater (Zainab et al. 2020), lakes (Yang et al. 2018), rivers (Singh et al. 2019), aquaculture farms (Chen et al. 2018a, b), and more recently even in coral reefs (Liu et al. 2020; Zhang et al. 2018, 2019).

According to EPA, USA (2008), coral reefs are the most diverse, productive, and complex ecosystems, inhabiting 25% of all marine life. They offer millions of ecological, economic, and societal benefits globally, generating a revenue of USD 11 trillion per annum (Rivera et al. 2020). However, in recent decades, coral reefs have been threatened due to increasing natural and anthropogenic stress, mainly from rising sea surface temperatures (SST)-induced coral bleaching, increasing levels of CO<sub>2</sub> leading to ocean acidification, ocean deoxygenation, coral diseases, unsustainable fishing practices, and land-based pollution (Hughes et al. 2018;

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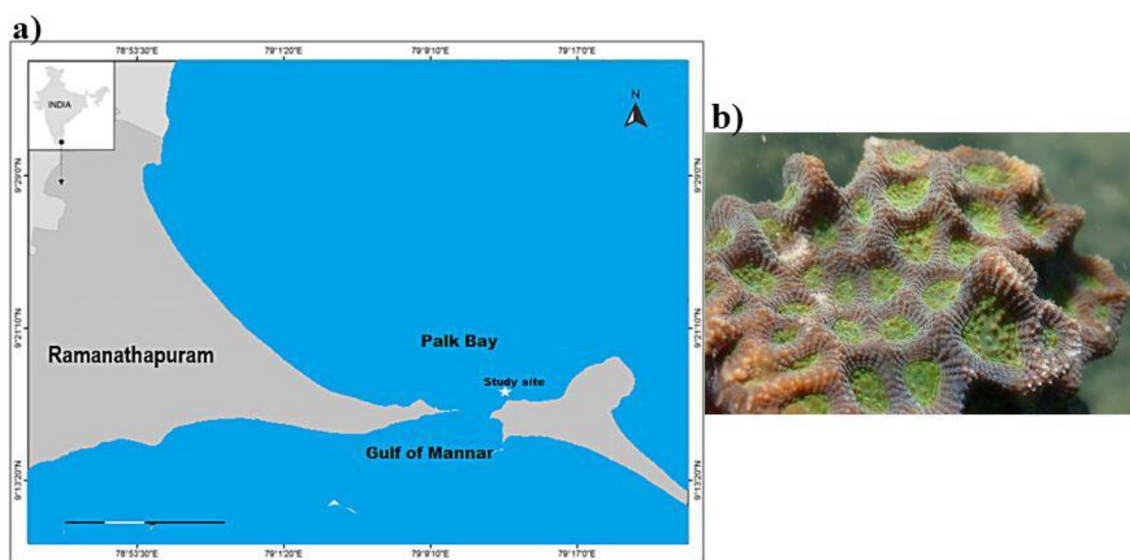
Hussain and Ingole 2020; Hussain et al. 2016; Burke et al. 2011). Increase in SST above the coral's threshold causes the breakdown of the symbiotic relationship between the coral host and symbiotic algae, triggering the expulsion of photosynthetic algal symbiont, leading to coral bleaching (Hughes et al. 2018; Hussain and Ingole, 2016). Additionally, the increase in anthropogenically induced CO<sub>2</sub> has reduced seawater acidity by 0.1 pH unit (30% increase in acidity) (IPCC, 2007). An increase in seawater acidity reduces the carbonate saturation levels and depletes the carbonate available for calcification process, leading to dissolution of calcium carbonate skeleton (aragonite mineral) of corals and other calcifying reef organisms (Shetye et al. 2020), thus weakening the corals. According to recent research, loss of oxygen (O<sub>2</sub>) from oceans, due to combined effects of anthropogenic nutrient overload and global warming (called ocean deoxygenation) is a bigger and more imminent threat to the survival of corals as compared to ocean acidification and thermal stress (Hughes et al. 2020). Decline in corals due to the above-mentioned threats have long-term social, ecological, and economic impacts, impacting the livelihood of millions of people globally (Burke et al. 2011).

The intricate and delicate network of interactions between the coral host, symbiotic *Symbiodiniaceae* (photosynthetic algae), and numerous symbiotic coral holobiont members such as bacteria, fungi, viruses, archaea, and endolithic algae is crucial to the survivability of corals (Glasl et al. 2016; Rohwer et al. 2002; Olson et al. 2009; Ritchie 2006). The stability of this coral holobiont may be altered by changes in the balance of the microbial community associated with corals as well as its functional ecology would change under the strain of climate change combined with

the rise in antibiotic contamination in coral reefs. This would lead to disturbances in ecological services like biogeochemical cycling, nitrogen fixation, nutrient availability, and coral biomineralization (Zhao et al. 2019).

Palk Bay located on the south-east coast of India is a near-shore, unprotected shallow fringing reef (Fig. 1a) with an average depth of 3 m situated between the latitude of 9°17' N and longitudes 79°17'40" E and 79°8' E (Thinesh et al. 2019). According to reports, Palk Bay's coral reef ecosystem is home to 66 species of corals from 23 different coral taxa (Ramesh et al. 1996; Thinesh et al. 2019). Palk Bay (PB), which separates the northern Sri Lankan coast from the coast of south-east India, is a zone of intermixing waters that opens into both the Arabian Sea (AS) and the Bay of Bengal (BoB) (Sneha et al. 2016). The area is also a major fishing harbour and a popular pilgrimage site located in Rameswaram (Meera 2016; James et al. 2021). Additionally, the Bay serves as a sink for anthropogenic effluents as it is where three rivers Vaishali, Vaigai, and Valryar discharge their water into the sea.

Recent studies have reported a major decline in coral cover in Palk Bay (Edward et al. 2015; Marimuthu et al. 2020), possibly due to the combined effects of recurrent coral bleaching events (Thinesh et al. 2019), *Terpios* sponge infestation (Thinesh et al. 2015), coral diseases like Black Band disease, *Acropora* white syndrome, white band disease and skeletal growth anomalies (Thinesh et al. 2011, 2014, 2017; Rajasabapathy et al. 2020), microbial pathogens (Meera 2016), and local anthropogenic stress. In the past few decades, Palk Bay has been exposed to heavy anthropogenic disturbances such as sewage and industrial discharge, oil pollution, trap fishing and fish processing



**Fig. 1** a Map of study site in Palk Bay and b *Favites abdita* coral colony from the sampling site

industries (Ramesh 1996; Thinesh et al. 2019). Over recent years, a high amount of heavy metal contamination has been recorded from marine sediments and seaweeds of Palk Bay, indicating an anthropogenic source of heavy metal pollution (Perumal et al. 2021; Rajaram et al. 2020). James et al. (2021) reported a high abundance and diversity of microplastics from surface water, marine sediment, and fish gut collected from Palk Bay, indicating anthropogenic stress in Palk Bay. Additionally, Meera (2016) observed high concentrations of pollution indicator bacterial groups (total *Coliforms* and total *Streptococci*) from marine sediment and water samples, linked to fecal pollution in coastal waters of Palk Bay. Therefore, it becomes imperative to monitor the increase and spread of antibiotic-resistant bacteria in a site so anthropogenically stressed as Palk Bay.

Recent advancements in whole genome sequencing (WGS) have opened new avenues to study resistance mechanisms in organisms. WGS coupled with powerful bioinformatic tools can provide information on the antibiotic resistance, pathogenic and virulence genes present within an organism, at reduced cost and time (Reuter et al. 2015). Therefore, the present study aims to isolate antibiotic-resistant bacteria from the anthropogenically disturbed reef of Palk Bay, followed by whole genome sequencing. This will provide key insights into the genome of the isolate and the presence of antibiotic resistance genes.

## Methodology

### Sampling site and sample collection

The sampling site was located near the eastern side of Pamban bridge at 9° 17' 46.38" N and 79° 13' 26.21" E coordinates (Fig. 1a). Triplicate coral fragments from three *Favites abdita* colonies at a depth of 2 m, separated by approximately 50 m from each other were collected by skin diving and snorkeling. We sampled *F. abdita* (Fig. 1b), an encrusting massive coral species as it is one of the dominant species in the study area (Sukumaran et al. 2011). The samples were kept in whirl pack bags in seawater and brought to the laboratory in ice packs.

### Symbiodiniaceae (algal cell) extraction and bacterial isolation

Each coral fragment was placed in a ziploc bag, sealed in all but one corner. Agaro® waterpik filled with 0.22 µm filtered and autoclaved seawater was inserted into an open corner of ziploc and the waterpik was turned on, flushing the tissue out from the coral skeleton. The collected tissue extracted was homogenized using mortar and pestle and placed into 50 ml conical tubes. The homogenized extract was centrifuged at

6000 g for 3 min using HeroLab Uni Cen MR® centrifuge. The supernatant was discarded carefully. 40 ml fresh autoclaved seawater was added to re-suspend algae. Washing was repeated (4–5 times) until a thick brown color, combining tube volumes. 100 µl extract was serially diluted and plated on Zobell Marine agar. The plates were incubated at 28 °C and the colonies were enumerated and morphological analysis was conducted after 48 h of incubation.

### Morphological analysis and biochemical assay

Conventional microbiological tests were performed on the bacterial strain. The tests included Gram staining, IMViC, oxidase and catalase test (Bergey 1994; McDevitt 2009; MacWilliams 2009). The bacterium was observed under a light microscope, and Gram staining was performed to assess the morphology. The motility, indole production, and sulfur reduction of the bacteria were determined by the SIM agar test (MacWilliams 2012). Urea, nitrate, and gelatin utilization tests were also performed following Bergey (1994).

### Antimicrobial resistance profiling

The isolated strain was tested for multidrug resistance profiling. The susceptibility test was carried out using the Kirby–Bauer disk-diffusion method (Hudzicki 2009) following CLSI/EUCAST (2012, 2022) criteria. The test antibiotic drugs included gentamycin (10 mcg), amikacin (30 mcg), tobramycin (10 mcg), ciprofloxacin (5 mcg), levofloxacin (5 mcg), cefepime (50 mcg), ceftazidime (10 mcg), penicillin (10 mcg), ampicillin (25 mcg), meropenem (10 mcg), erythromycin (5 mcg), clindamycin (10 mcg), vancomycin (30 mcg), tetracycline (30 mcg), and polymyxin B (300 U) (Behera et al. 2010).

### DNA extraction

Bacterial DNA was isolated using the DNeasy Plant Mini kit (Qiagen, Germany) following the manufacturer's protocol. The extracted DNA was observed on 0.8% agarose gel, and concentration was measured using Thermo Fischer nanodrop equipment. 16S rRNA gene sequence (Accession No: OR378794) was used to confirm the identity of the isolated bacterial strain using primers 8F 5'-AGAGTTTGATCC TGGCTCAG-3' and 1492R (5'-GGTTACCTTGTTACG ACTT-3') (Galkiewicz and Kellogg 2008). The species was confirmed using EzBioCloud (Yoon et al. 2017).

### Library preparation and QC for whole genome sequencing (WGS)

Libraries for WGS were prepared using the TruSeq DNA Nano LP kit (Illumina #15041877, Illumina #20665713)

according to manufacturer's instruction. The library insert size was determined using TapeStation 4150 (Agilent) utilizing high-sensitive D1000 screentapes (Agilent # 5067-5582) following the manufacturer's protocol. Final libraries were quantified using Qubit 4 Fluorometer (ThermoFisher #Q33238) using a DNA HS assay kit (ThermoFisher #Q32851) following the manufacturer's protocol. Sequencing was performed on the Illumina (NOVASEQ 6000) with paired-end at Molsys Pvt Ltd. Sequencing facility, Bangalore, India.

### Genome quality control

The raw sequence was preprocessed for filtering low-quality reads and trimming the adapter content using the Fastp tool (Chen et al. 2018a, b). The processed reads were assembled using megahit v1.2.9 with default parameters. Quality control of the genome assemblies was evaluated using QUAST v5.0.2 (Gurevich et al. 2013). Ragtag was used for scaffolding based on the reference genome of *Pseudomonas stutzeri*. Subsequently, quality control using CheckM v1.1.3 (Parks et al. 2015) was performed to check the genome completeness.

### Genome annotation and assembly

The annotation of the draft genome was performed using BAKTA (Schwengers et al. 2021) tool which comprises the following functional and domain databases as AMRFinder-Plus (Feldgarden et al. 2019), COG (Galperin et al. 2021), DoriC (Luo and Gao 2019), ISFinder (Kamoun et al. 2013), MOB-suite (Robertson et al. 2020), Pfam (El-Gebali et al. 2019), RefSeq (Li et al. 2021), Rfam (Kalvari et al. 2021), UniProtKB/Swiss-Prot (UniProt Consortium 2021), VFDB (Liu et al. 2019). All annotation types such as COG, GO, KEGG pathways, Enzyme class, and CDS were predicted.

### Prediction of AMR genes and genomic islands

The antimicrobial resistance (AMR) genes were predicted from CARD 2023 database (Alcock et al. 2023). The predicted genomic features were represented in a circular view using the Proksee tool (<https://proksee.ca/>). Genomic islands (GIs) prediction was carried out for our target isolate *S. frequens* CAM01 using the IslandViewer 4 (Bertelli et al. 2017).

### Dataset preparation and phylogenomic analysis

The genome sequence data of *S. frequens* CAM01 were uploaded to the Type (Strain) Genome Server (TYGS), a whole-genome-based taxonomic analysis (Meier-Kolthoff and Göker 2019). The analysis also made use of recently

introduced methodological updates and features (Meier-Kolthoff et al. 2022). The results were provided based on (i) the determination of closely related type strains using the genome BLAST distance phylogeny method (GBDP) with the algorithm "coverage" and distance formula  $d_5$  to determine exact distances for each user genome (Meier-Kolthoff et al. 2022), and (ii) the pairwise comparison of genome sequences, to predict the intergenomic distance among the selected type strains. Digital DDH (dDDH) values and confidence intervals were computed using the GGDC 3.0 (Meier-Kolthoff et al. 2013). The constructed phylogenomic tree was edited and designed in MEGA X (Kumar et al. 2018) and iTOL v6 (Letunic and Bork 2007).

### Prediction of metabolic genes

The genes of our target isolate were predicted based on the gene ontology (GO) term from the reference *Pseudomonas* genus. The predicted genes were further imported into the STRING v11.5 (Szklarczyk et al. 2017) server against the reference organism as "*Pseudomonas stutzeri* A1501" to perform the draft PPINs. The full STRING network type was selected from PPINs, to identify both functional and physical protein associations. All seven methods of interaction sources "Textmining, Neighborhood, Experiment, Gene fusion, Database cross search, Co-occurrence and Co-expression" were selected before performing the analysis. We fix the network with a "High confidence" score as 0.7 (70% of similarity) is the minimum required interaction. Based on the GO annotation, the genes involved in respective KEGG pathways were predicted. The final network was designed using Cytoscape v3.9.1 (Shannon et al. 2003) and validated using a Network analyzer module (Doncheva et al. 2019).

### Prediction of secondary metabolites

The assembled genome of our isolate was further imported into the antiSMASH v7.0 (Blin et al. 2021) server to predict secondary metabolites using a "strict" detection (well-defined clusters containing all required parts). All structural and domain databases were selected for analysis to predict the metabolites.

## Results

Morphologically, *S. frequens* CAM01 was characterized as Gram-ve, rod-shaped bacteria, 1–3  $\mu\text{m}$  in length, 0.5  $\mu\text{m}$  in width, and the colony appeared orangish brown with a wrinkled appearance. Biochemically they tested positive for the characteristic tests of *Stutzerimonas* (Supplementary Word File: Table S1).



## Antimicrobial resistance profiling

Antimicrobial resistance profiling of the isolate revealed resistance against several clinically important antibiotics. The isolate showed resistance against 73% of the antibiotics tested including the antibiotics belonging to classes Fluoroquinolones, Cephalosporin, Penicillin, Macrolides, Lincosamides, Glycopeptides, and Tetracyclines. (Table 1).

## Genomic characterization of *Stutzerimonas frequens* CAM01

The draft genome sequence of *S. frequens* CAM01 was submitted to the NCBI biological database with corresponding accession PRJNA916631. It contains 4.67 Mbp with 52 contigs and its genome coverage as  $\approx 100\times$ . The largest contig base pair length is 3642853 bp. The genome consists of 64.05% of G + C content. The paired-end sequence mean read length of 150 bp generated approximately 28.9 million reads and after trimming based on the quality filter ( $Q > 30$ ), approximately 28.01 million reads were assembled using megahit. The genome sequence was predicted to comprise 3,847 encoding genes, of which 3,739 were involved in protein-coding genes (CDSs), rRNA genes (5), and tRNA genes (53) (Fig. 2; Supplementary Excel 1). In addition, 24 genes were encoded for ncRNA regions. Overall, 1,328 proteins are found to be unknown function, putative, uncharacterized, and hypothetical proteins (Table 2). The predicted circular view represents the genomic context of each ORF of *S. frequens* CAM01.

**Table 1** Phenotypic antibiotic resistance profiling of isolate *S. frequens* CAM01

| Antibiotic class | Antibiotic    | Zone of inhibition (mm) |
|------------------|---------------|-------------------------|
| Aminoglycosides  | Gentamycin    | 14 (R)                  |
|                  | Amikacin      | 13 (R)                  |
|                  | Tobramycin    | 21 (S)                  |
| Fluoroquinolones | Ciprofloxacin | 18 (R)                  |
|                  | Levofloxacin  | 17 (R)                  |
| Cephalosporin    | Cefepime      | 0 (R)                   |
|                  | Ceftazidime   | 26 (I)                  |
| Penicillin       | Penicillin    | 0 (R)                   |
|                  | Ampicillin    | 0 (R)                   |
| Carbapenems      | Meropenem     | 25 (S)                  |
| Macrolides       | Erythromycin  | 0 (R)                   |
| Lincosamides     | Clindamycin   | 0 (R)                   |
| Glycopeptides    | Vancomycin    | 0 (R)                   |
| Tetracyclines    | Tetracycline  | 3 (R)                   |
| Lipopeptide      | Polymyxin B   | 15 (S)                  |

## Analysis of phylogenomic imprints

The phylogenomic tree was inferred from GBDP distances using the FastME 2.1.6.1 (Lefort et al. 2015) with 100 pseudo-bootstrapping replicates. The genomic features of each selected type strain for phylogenetic tree construction were represented in Supplementary Excel 2. The phylogenomic tree consisted of 17 complete genomes, 1 belonging to our target and the other 16 being the closest type strains. The phylogenomic tree inferred that our target *S. frequens* CAM01 genome is closest to type strain *S. frequens* DNSP21 (Fig. 3). The genome sequences of *Stutzerimonas stutzeri* are 96.945% identical by average nucleotide identity (ANI) to the type genome of *Stutzerimonas frequens*, with 91.4% coverage of the genome. It was further confirmed by pairwise genomic comparisons based on digital DNA–DNA hybridization (dDDH) values with the type strain showed  $\sim 88\%$  of similarity, as represented in Supplementary file Table S2.

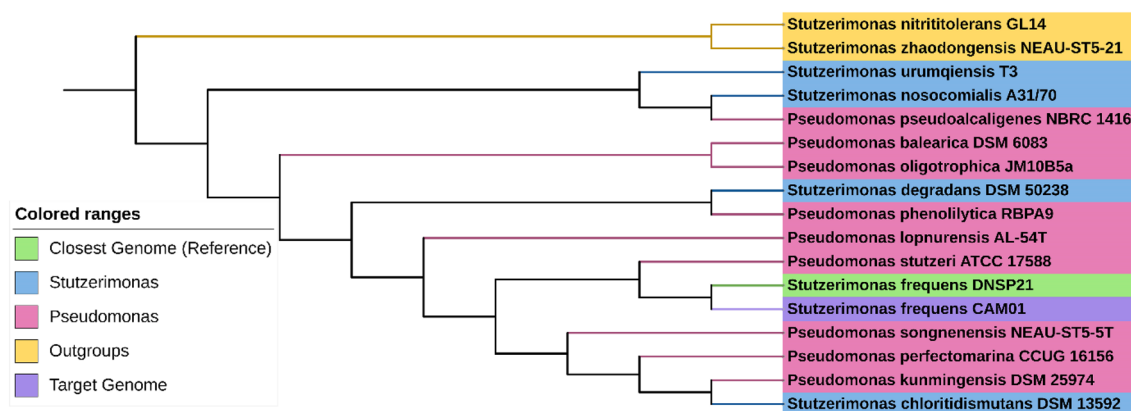
## Antibiotic resistance determinants

Interestingly, four antimicrobial resistance genes were predicted from the CARD database, namely, *adeF*, *rsmA* APH(3'')-Ib, and APH(6)-Id encoding resistance–nodulation–cell division (RND) antibiotic efflux pump and aminoglycoside phosphotransferase. The AMR genes mined and their mode of action can be seen in Table 3. The genes with evidence of horizontal origins were identified based on genomic islands prediction, also we identified one of the antimicrobial resistance gene APH(3'')-Ib was present in them, as represented in Supplementary Table S3.

## Annotation of COG functional classification

Using COG, the functional classification of each gene within a genome was predicted. A maximum number of genes were found to be involved in functions like metabolism (41%), cellular processing and signaling (27%), whereas 21% of genes were involved in information storage and processing. Additionally, 11% of genes (232) were poorly characterized, as depicted in Fig. 4a. Thus, it concludes, 89% of genes in *S. frequens* CAM01 are thoroughly annotated and functional role is determined. The individual functional profile reveals that the maximum number of genes are involved in amino acid transport and metabolism (208), translation, and biogenesis process (203), as depicted in Fig. 4b. Carbohydrate (111), lipid (127) and coenzyme transport (111), transcription (127), replication (94), and cell wall biogenesis (141) are the second highest number of genes involved in the COG functional classes. The classes C, D, F, N, Q, U, and V have shown less than 50 genes.





**Fig. 3** Representing the phylogenomic reconstruction among *S. frequens* CAM01 and selected closest type strain genomes. The species in each node and branch are grouped and colored according to their genome

**Table 3** Antibiotic resistance genes and their mechanism of action predicted in *Stutzerimonas frequens* CAM01 (<https://card.mcmaster.ca>)

| Resistance gene    | AMR gene family                                                  | Drug class                                                                     | Resistance mechanism    |
|--------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------|
| <i>adeF</i>        | Resistance–nodulation–cell division (RND) antibiotic efflux pump | Tetracycline antibiotic, fluoroquinolone antibiotic                            | Antibiotic efflux       |
| <i>rsmA</i>        | Resistance–nodulation–cell division (RND) antibiotic efflux pump | Diaminopyrimidine antibiotic, phenicol antibiotic, fluoro-quinolone antibiotic | Antibiotic efflux       |
| <i>APH(3'')-Ib</i> | APH(3'')                                                         | Aminoglycoside antibiotic                                                      | Antibiotic inactivation |
| <i>APH(6)-Id</i>   | APH(6)                                                           | Aminoglycoside antibiotic                                                      | Antibiotic inactivation |

*R* resistant, *S* susceptible, *I* intermediate

in metabolic reactions (Piek et al. 2014; Paul et al. 2019). Other enzymes such as lyases shared (65 enzymes) isomerases (23), ligases (44), and translocases (14), as depicted in Fig. 6.

### Analysis of PPINs based on the KEGG pathway

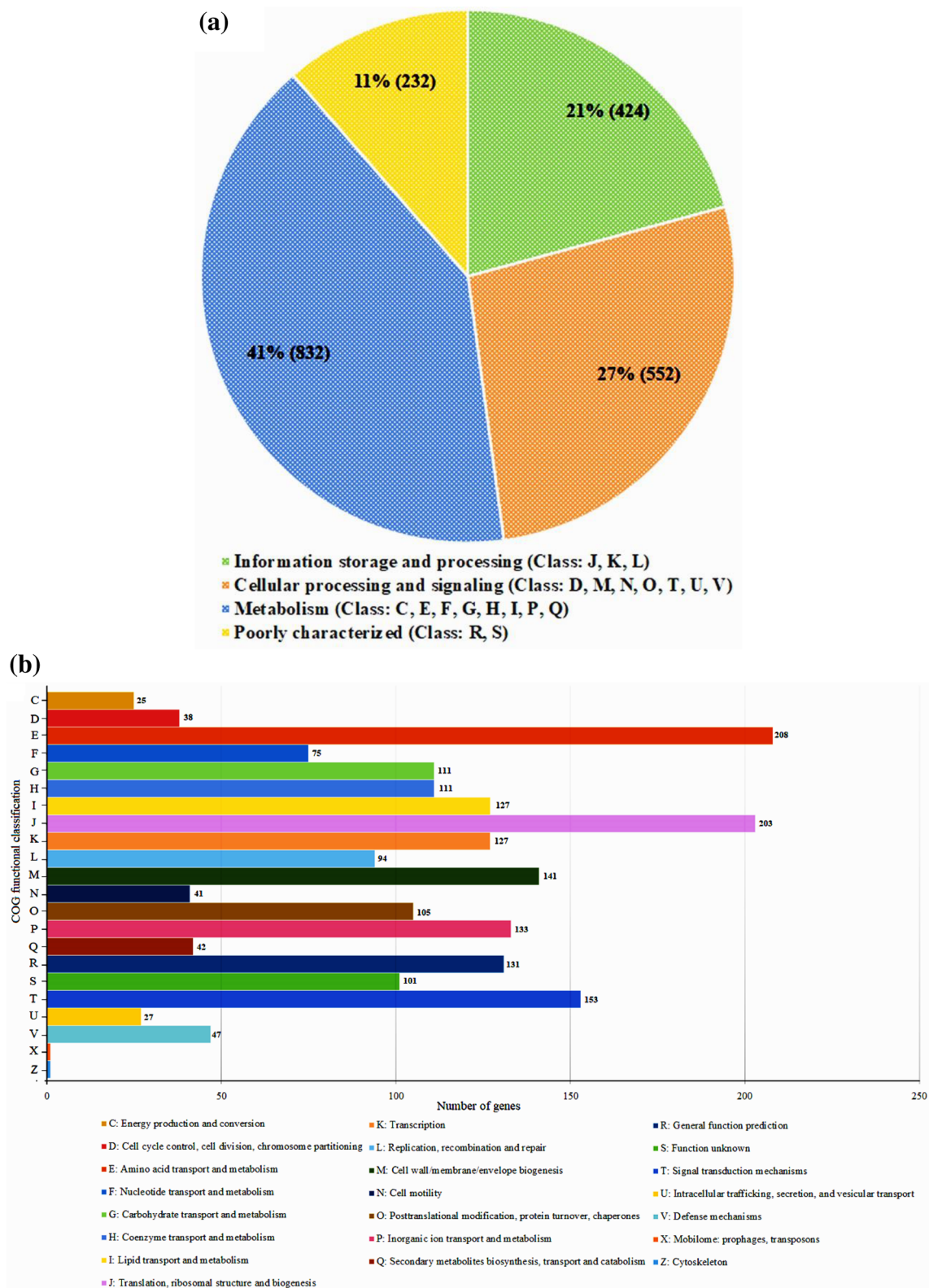
In the present study, we performed the proteome-wide protein interaction networks (PPINs) based on their KEGG pathway, as depicted in Fig. 7. Overall, 1829 genes were predicted based on GO term, involved in 19 essential metabolic pathways. Each pathway was represented in different color legends. A maximum number of genes (468) are involved in the metabolic process such as secondary metabolites biosynthesis (200) and microbial metabolism in diverse environments (128). Interestingly we found that 83 genes are involved in amino acid biosynthesis, and nucleotide metabolism (78), including purine (52) and pyrimidine metabolism (26). The second highest number of genes (343) were found in general metabolism, single reaction, and non-pathway, whereas ribosome metabolism contains 47 genes, which play a function in RNA modification and processing (Gilbert 1986). Moreover, 30 genes involved in

Flagellar assembly and 17 in protein export metabolism. The amino acids such as alanine, aspartate, and glutamate metabolism contain 20 genes, which play a crucial part in the intermediate metabolism of amino acids and glucose (Sookoian and Pirola 2012), as well as metabolic mediators of host–pathogen interactions (Ren et al. 2018). As per the previous investigation by Kellock et al. (2020), aspartate amino acid plays a role in preventing coral calcification as a result of ocean acidification. Overall, 17 and 59 genes are involved in energetic metabolism (TCA cycle) and carbon metabolism. The entire PPINs consisted of 986 nodes and 16,587 edges, perpendicular to each other. The PPI-enriched *p* values shown < 1.0e-16, it determines the stability of nodes and edges in PPINs (Fig. 7).

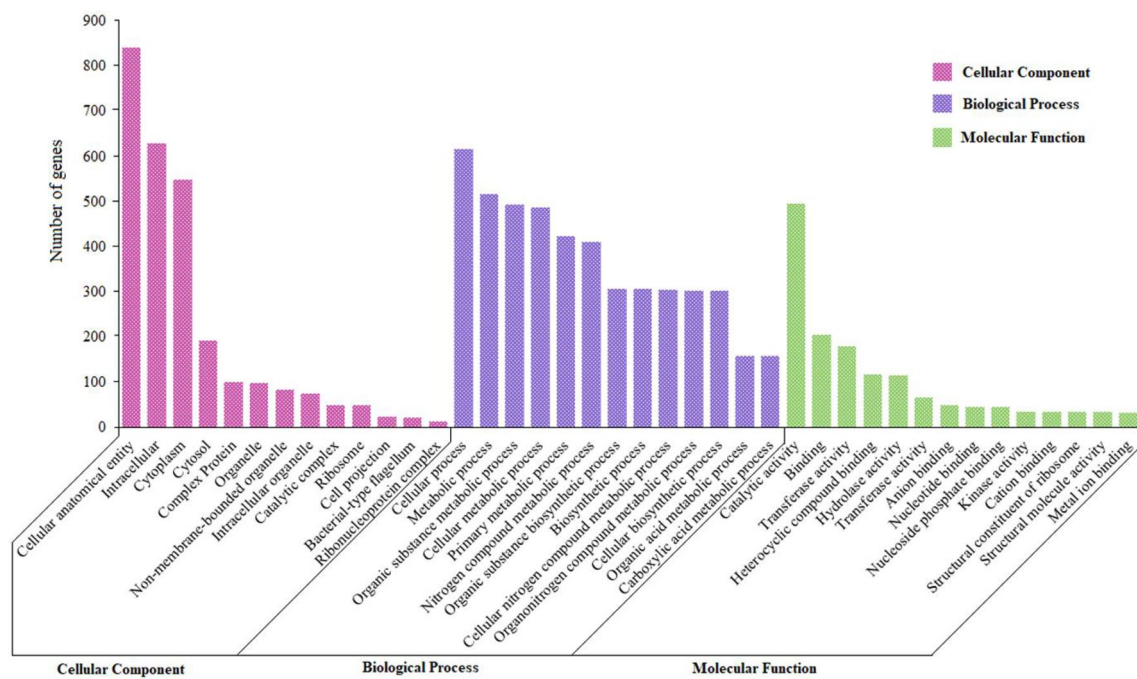
### Secondary metabolites

Interestingly we found four secondary metabolites in various genomic positions. Such metabolites include the Aryl polyene cluster found in the genomic location as 42,696 to 86,295, sharing 45% similarity with the most known cluster in *Aliivibrio fischeri* ES114, a marine bacterium. Whereas other metabolites like NRPS-independent



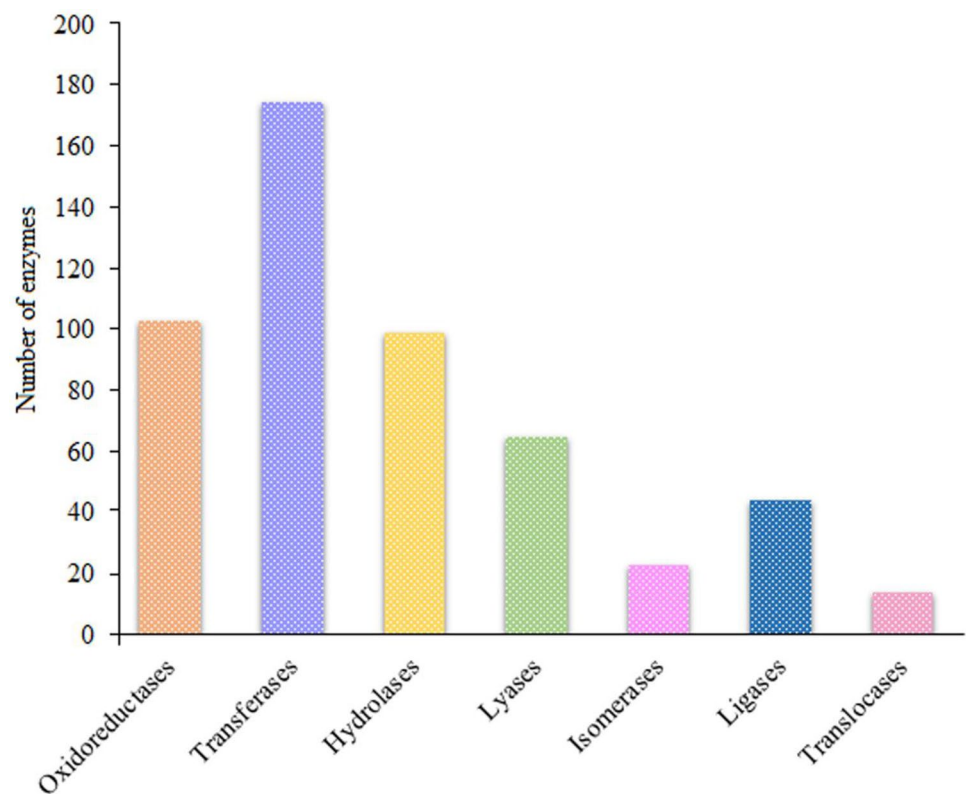


**Fig. 4** Predicted COG functional classification in percentage (a) and in numbers of individuals (b) of *S. frequens* CAM01

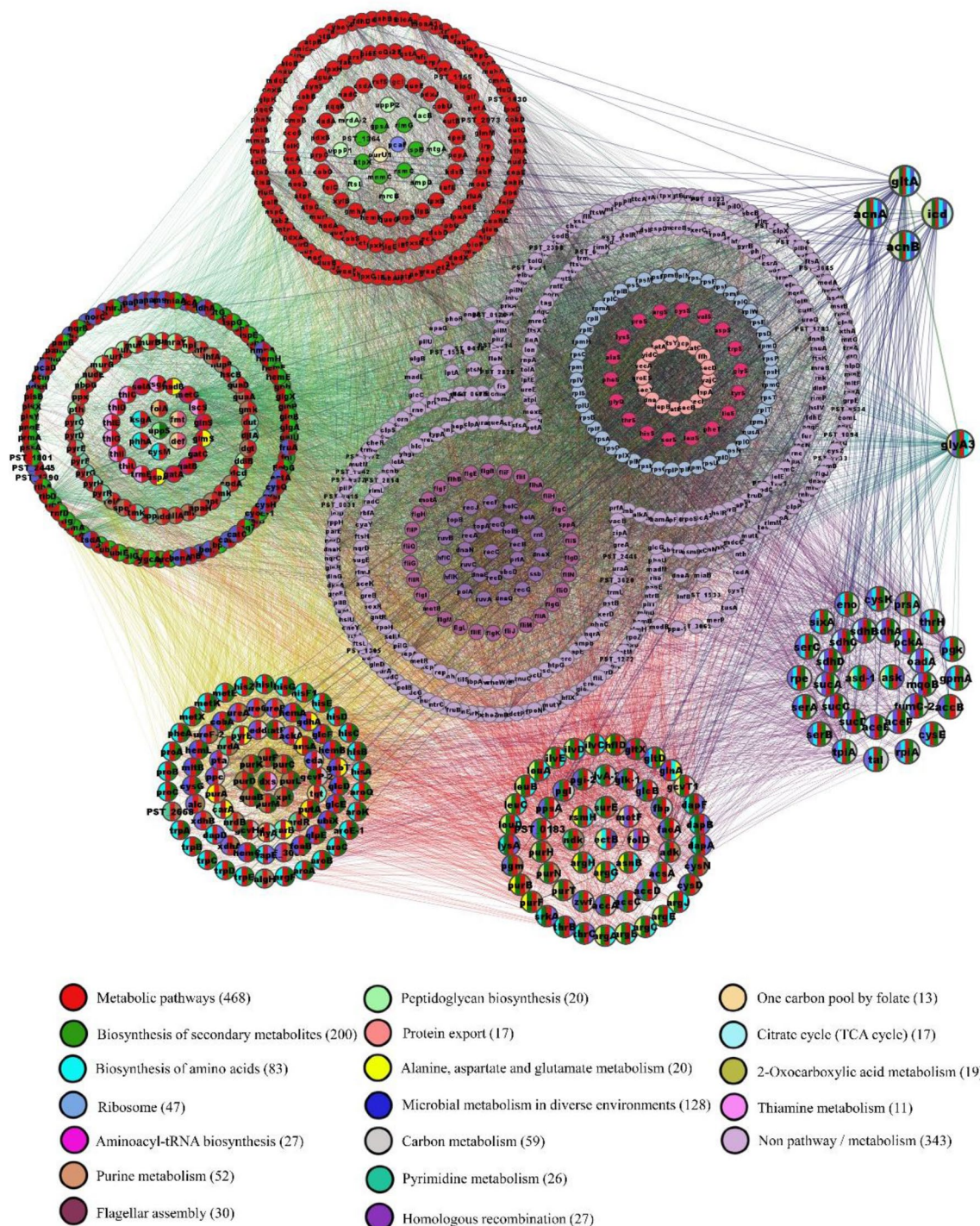


**Fig. 5** The biological processes, cellular components, and molecular function of predicted genes based on the gene ontology of *S. frequens* CAM01

**Fig. 6** Predicted percentage of enzyme classification of *S. frequens* CAM01 genome







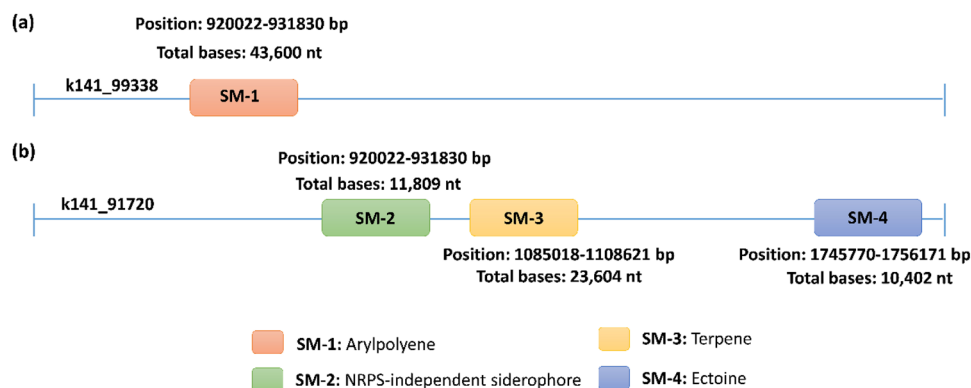
**Fig. 7** Predicted proteome-wide protein interaction network and network properties of our target genomic isolate *S. frequens* CAM01 against reference *P. stutzeri* A1501 genome

siderophore (920,022 to 931,830), terpene (1,085,018 to 1,108,621), and ectoine (1,745,770 to 1,756,171) share 83–100% similarity with a reference genome, as depicted in Fig. 8; Supplementary Figure S1.

## Discussion

Several recent studies have highlighted the rise of anthropogenic marine pollution in Palk Bay, leading to the loss of coral cover and biodiversity in ecologically sensitive

**Fig. 8** Predicted secondary metabolites **a** contig k141\_99338, and **b** contig k141\_91720 using Anti-smash server



reefs (Edward et al. 2015; Marimuthu et al. 2020; Manikandan et al. 2014). As a result of poor water quality, corals in Palk Bay have been plagued with coral diseases and coral-eating sponge (*Terpios*) infestation in recent years (Thinesh et al. 2014, 2015, 2017; Rajasabapathy et al. 2020). The study area is under immense anthropogenic stress being a major fishing harbour and a famous pilgrimage site, receiving hundreds of thousands of pilgrims every day (Meera 2016; Jeyasanta et al. 2020; James et al. 2021). Additionally, the shore near the sampling site is also an open defecation area (pers. observation), polluting the reef area with fecal contamination. This has been corroborated by the work of Meera (2016), who observed high values of pollution index ( $> 1$ ) across sites in Palk Bay, indicating fecal pollution. The study also reported a high amount of fecal-matter-associated microbes, which would make the water unsuitable for recreational activities. Interestingly, in accordance with our findings, Meera (2016) proposed that the fecal pellet may be a carrier of antibiotics and may lead to antibiotic pressure in microbial strain in the study area. In agreement with our study, Sneha et al. (2016) reported widespread distribution of multiple antibiotic-resistant *Vibrio* spp. across Palk Bay, indicating an anthropogenic input for the rise of AMR. Furthermore, high concentration of heavy metals, particularly cadmium (Cd), iron (Fe), and manganese (Mn) has been reported from the marine sediments of Palk Bay (Perumal et al. 2021; Rajaram et al. 2020). High contamination of Cd poses a high ecological risk and points toward anthropogenic inputs like a discharge of municipal waste, domestic discharge, industrial waste, fishing harbour-related activities, and agricultural run-off (Perumal et al. 2021). High concentrations of Fe and Mn as observed in the same study are a result of various anthropogenic activities like localized oil pollution, the use of antifouling and corrosive paints in boats, dredging and landfilling.

With the rising threat of increasing SST coupled with anthropogenic stress, Palk Bay can be a hotspot for the emergence of antibiotic-resistant bacteria. Therefore, this study

provides genomic insights of one such antibiotic-resistant bacteria isolated from coral species, *Favites abdita* of Palk Bay.

### Phylogenomic and genomic insights

*Stutzerimonas* is a recently proposed genus within the family Pseudomonadaceae, which was formerly composed of the strains belonging to the phylogenetic group of *Pseudomonas stutzeri* species complex (Lalucat et al. 2021; Gomila et al. 2022). *Stutzerimonas stutzeri* is the type species of the *Stutzerimonas* genera, consisting of 22 different genomovars, which are defined as genotypically similar groups that cannot be differentiated phenotypically (Gomila et al. 2022). Using phylogenomics, which is the study of the origin, replication, and ancestral relationship among selected species or groups of taxa (Chellapandi et al. 2018; Murugan et al. 2020; Prathiviraj and Chellapandi 2020a, 2020b), we inferred that our isolate *S. frequens* CAM01 is closest to type strain *S. frequens* DNSP21. In agreement with our study where we isolated the strain from *Favites abdita*, members of *Stutzerimonas* gen. nov. have been isolated predominantly from wastewater, sludge, compost, oil-contaminated sand, soil, plants, marine water, hydrothermal vents, and marine sediments from intertidal shore to depths of 11,000 m (Gomila et al. 2022). The genome size of *Stutzerimonas* gen. nov. varies from 4.07 Mb to 5.05 Mb and GC content varies from 59.6 to 67.0% (Lalucat et al. 2021) and the draft genome size of *Stutzerimonas frequens* sp. nov. is 4.4 Mb with GC content of 58.7% (Gomila et al. 2022). This corroborates our findings, where we got a genome size of 4.406 Mb and a GC content of 64.05%.

*Stutzerimonas frequens* is a Gram-negative, denitrifying bacterium, widely dispersed in the marine environment (Gomila et al. 2022). The phenotypic and biochemical characteristics of our strain *Stutzerimonas frequens* CAM01 corroborate with the description of other strains of *Stutzerimonas frequens* by Gomila et al. (2022).



## Antimicrobial resistance insights

Following antibiotic resistance profiling, *S. frequens* CAM01 was characterized as a multidrug-resistant (MDR) bacterium. It showed resistance to several clinically important antibiotics, in agreement with previous studies on antibiotic resistance profiling of *Pseudomonas stutzeri* species complex (Madaha et al. 2020; Tattawasart et al. 1999; Russell and Mills 1974; Kwatra 2021). Multidrug resistance points toward the presence of various mechanisms for antibiotic resistance in *S. frequens*. This was further confirmed following the genomic analysis of the resistance genes, which revealed the presence of AMR genes namely, *adeF*, *rsmA* APH (3'')-Ib, and APH (6)-Id. Our isolate showed resistance to antibiotics classes of fluoroquinolone and tetracycline, which is consistent with an antibiotic resistance gene hit of *adeF* (resistant to fluoroquinolone, tetracycline). The gene *adeF* is a membrane fusion protein of the multidrug efflux complex *AdeFGH* (Leus et al. 2018). The gene *adeF* encodes for a resistance–nodulation–cell division (RND) antibiotic efflux pump, which impairs resistance against fluoroquinolone and tetracycline (Reygaert (2018); Coyne et al. 2010; Alcock et al. 2023). The *adeF* is reported to be carried simultaneously by transposons, integrons, and plasmids across different hosts (Wang et al. 2022) and there is a risk of it being spread extensively through food chains owing to frequent horizontal gene transfer (Li et al. 2022).

The *rsmA* is also a member of the resistance–nodulation–cell division (RND) antibiotic efflux pump AMR gene family and is reported to regulate the virulence of several bacterial pathogens in both plants and animals (Ramírez-Zapata et al. 2020; Burrowes et al. 2005; Heurlier et al. 2004). It confers antibiotic resistance by directing the pumping of antibiotics out of a cell, a mechanism known as antibiotic efflux. Studies on *P. aeruginosa* have shown that *rsmA* negatively affects the expression of *MexEF-OprN* and other *Mex* pumps, conferring resistance to various antibiotics (Mulcahy et al. 2006). Linares et al. (2006) observed that *rsmA/rsmZ* provided resistance to *P. aeruginosa* against tobramycin, ciprofloxacin, and tetracycline. Thus, the presence of *rsmA* within *S. frequens* CAM01 genome could explain its resistance to ciprofloxacin and tetracycline. Interestingly, *S. frequens* CAM01 was moderately sensitive to tobramycin and its minimum inhibitory concentration against tobramycin must be further examined.

APH(3'')-Ib and APH(6)-Id encode for aminoglycoside phosphotransferase, associated with resistance to aminoglycosides, involved in streptomycin inactivation (Ashenafi et al. 2014; CARD, 2023). This could be the reason for resistance against aminoglycoside antibiotics like gentamycin and amikacin in *S. frequens* CAM01.

Resistance against other antibiotics like  $\beta$ -lactams (penicillin, ampicillin, cefepime) and macrolids (erythromycin)

could be possibly due to the presence of RND pumps coded by *adeF* and *rsmA* in our isolates *S. frequens* CAM01. In addition to a known mechanism like the presence of  $\beta$ -lactamase enzyme and mutation of  $\beta$ -lactam targets, methods like altering the permeability of the bacterial outer membrane or increasing the efflux pump activity may provide resistance against  $\beta$ -lactam antibiotics in Gram-negative bacteria (Lingzhi et al. 2018). An increase in the activity of efflux pumps can be performed by the *MexAB* and *OprM* antibiotic efflux pump, which is a major cause of resistance in Gram-negative bacteria (Wilke et al. 2005). As already established *rsmA* plays a role in the expression of *MexEF*, *OprN*, and other *Mex* pumps, conferring resistance to various antibiotics (Mulcahy et al. 2006).

Genomic islands were predicted from the present study to understand the evolution and horizontal transfer of genes from *Stutzerimonas* sp. The gene APH(3'')-Ib is associated with resistance mechanisms in many of the bacterial genomes and can evade the effects of antibiotics (Ashenafi et al. 2014). Thus, the genetic islands with aminoglycoside resistance gene may have a role in protecting the bacteria from various antibiotics and this agrees with our findings.

## Proteome-wide protein interaction networks

This was followed by proteome-wide protein interaction networks (PPINs) analysis which is the study to understand the mechanism of protein functions and the drug ability of macromolecules (Prathiviraj et al. 2019; Prathiviraj and Chellapandi 2020a, 2020b). It also helps us to identify the most essential hub genes (a protein that controls an entire subsystem) and KEGG pathways (Kiran et al. 2021; Sivakumar et al. 2021; Lakshmanan et al. 2022; Vinothkanna et al. 2022). Thus, our findings from PPINs predicted 1829 genes are found in the genome which plays a major role to unwind the genome characterization, cell physiology, and metabolism (Lv et al. 2015).

## Putative secondary metabolites

Secondary metabolites are a range of bioactive compounds with diverse chemical structures, produced by bacteria, fungi, plants, etc. (Katz and Baltz 2016). Secondary metabolites have been widely used in biotechnology and biomedical industries (Charlesworth and Burns 2015) owing to their antibacterial, antiviral, anticancer, anti-inflammatory, anti-malarial, and neuroprotective properties (Rocha et al. 2011; Bhanot et al. 2011). Microbial secondary metabolites provide its host bacteria with an array of benefits like protection from other bacteria by acting as growth inhibitors, quorum sensing molecules, metal quenchers, and transporting agents (Demain and Fang 2000; Wang and Lu 2017). Recent advancements in the field of bioinformatics have made it



possible to characterize secondary metabolite biosynthetic gene clusters (BGCs) in bacteria (Wang et al. 2019a, b). BGCs are defined as a local cluster of genes that encode a biosynthetic pathway, leading to the production of a secondary metabolite (Medema et al. 2015). Four secondary metabolites, namely Aryl polyene, NRPS-independent siderophore, terpenes, and ectoine were detected in our isolate.

It is speculated that the bacterial pigments of the aryl polyene type are structurally like the well-known carotenoids (Schöner et al. 2016) and just like the pigment carotenoids, may be acting as antioxidants. Interestingly, derivatives of aromatic polyene isolated from marine organisms have exhibited antibacterial and antitumor activities (Kwon et al. 2006). Terpenes of microbial origin are known to exhibit antimicrobial and antioxidant activity, acting as metabolites of microbial protection (Song et al. 2015; Zhao et al. 2008; Shindo et al. 2007).

Ectoines are compatible solutes that are used by microorganisms for protection against various stressors (Gunde-Cimerman et al. 2018) like osmotic stress (Czech et al. 2018), thermal stress (both high and low temperature, Salvador et al. 2018), desiccation stress (Tanne et al. 2014). They also serve to protect the lipid bilayer of the cell membranes, protects DNA from harmful ionizing radiation, and act as hydroxyl radical scavenger (Schröter et al. 2017; Brands et al. 2019). The presence of an NRPS-independent siderophore pathway indicates the synthesis of siderophores. Siderophores are used by bacteria for chelating ferric ions, an oxidative state of essential growth nutrient, iron (Kramer et al. 2020). Siderophores can perform other functions like protection from oxidative stress (Jin et al. 2018) and provide defence against other bacteria (Sassone-Corsi et al. 2016). Interestingly, the presence of antioxidant-producing bacteria in a heat-stressed coral indicates *S. frequens* role against photooxidation in the high thermal and UV-stressed reef of Palk Bay.

## Conclusion

This study focuses on the characterization of an antibiotic-resistant strain of *S. frequens* CAM01, isolated from the coral of the near-shore, anthropogenically polluted reef of Palk Bay, India. The whole genome sequencing approach was used to provide deeper insights into the genome of *S. frequens* CAM01. Phenotypic antibiotic resistance and the presence of several AMR genes indicate the potential ecological and public health implications of *S. frequens* CAM01 in the reef. As these AMR genes are located on mobile genetic elements like transposons, there is a risk of being carried to other bacteria through horizontal transfer, enhancing the prevalence of antibiotic-resistant bacteria in the reef environment. Additionally, the presence of

secondary metabolites may help the bacteria in its defence and virulence. Owing to the critical importance of the functional coral microbiome and the risk of community shift, it is imperative to study the presence of other antibiotic-resistant bacteria in Palk Bay's coral reef and to identify and regulate the source of antibiotic pollution.

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**Data availability** All nucleotide sequences assembled in this study have been deposited in the DDBJ/ENA/GenBank: *Stutzerimonas frequens* CAM01 assembly (accession code JAQBSX000000000) under the BioProject PRJNA916631.

## Declarations

**Conflict of interest** The authors declare no competing interests.

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